

Name: _____

Homework 8 | Math 341 | Cruz Godar

Due Wednesday of Week 9 at the start of class

Complete the following problems and submit them as a pdf to Canvas. 8 points are awarded for thoroughly attempting every problem, and I'll select three problems to grade on correctness for 4 points each. Enough work should be shown that there is no question about the mathematical process used to obtain your answers.

Section 9

In problems 1–3, find the change of basis matrix B that converts from the basis $\mathcal{B}$ for the vector space V to the standard basis, and use it to find $[\vec{v}]_{\mathcal{B}}$ for the given vector $\vec{v}$.

1. V is the subspace of $\mathbb{R}[x]$ of polynomials with degree at most 3, $\mathcal{B} = \{1, x + x^2 + x^3, x^3 - x, x^3 + 2x^2\}$, and $\vec{v} = 2 + 7x - x^2$.
2. $V = M_{2 \times 2}(\mathbb{R})$,

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\},$$

$$\text{and } \vec{v} = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}.$$

3. $V = \mathcal{L}(\mathbb{R}^4, \mathbb{R})$, $\mathcal{B} = \{T_1, T_2, T_3, T_4\}$, where

$$T_1 \left(\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \right) = 2x + y - z \quad T_2 \left(\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \right) = x - w \quad T_3 \left(\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \right) = y + z \quad T_4 \left(\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \right) = x + y + z + w,$$

and $\vec{v} : \mathbb{R}^4 \rightarrow \mathbb{R}$ is defined by $\vec{v} \left(\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \right) = 2x - 2y + 4z + 2w$.

In problems 4–6, find bases for V , $\ker T$ and $\text{image } T$, and verify that the Fundamental Theorem of Linear Algebra correctly relates the three.

4. $T : V \rightarrow \mathbb{R}$, where V is the subspace of $\mathbb{R}[x]$ of polynomials with degree at most 3, is defined by $T(a + bx + cx^2 + dx^3) = d - c$.

5. $T : M_{2 \times 3}(\mathbb{R}) \rightarrow \mathbb{R}^3$ is defined by

$$T \left(\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \right) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

6. $T : \mathbb{R}^2 \rightarrow \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ is defined by $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = S_{x,y}$, where

$$\begin{aligned} S_{x,y} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) &= \begin{bmatrix} x - y \\ 0 \end{bmatrix} \\ S_{x,y} \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) &= \begin{bmatrix} 0 \\ y - x \end{bmatrix}. \end{aligned}$$